

Course Code	Course Title	L	R	P	Cr
	AI Assisted Coding and Problem Solving using Python				
Program Core					

Product 1 – AI-Powered Student Utility Calculator

Day 1

Lab 1:

- Environment Setup & Python Fundamentals
- Configure Python & VS Code
- Setup AI Coding Assistants
- Variables & Input/Output
- Basic Calculator

Equipments/components/software required

- Personal Computer
- Python 3.14.6
- Visual Studio Code
- GitHub Copilot
- ChatGPT/Gemini
- Internet Connectivity

Curiosity/Context

How can Artificial Intelligence assist beginners in setting up a Python development environment and writing their first Python program with minimal effort?

Python Environment Setup

Activity 1: Install Python

- Download Python from the official website.
- Install Python.
- Configure PATH environment variable.
- Verify installation.

Commands

```
python --version  
pip --version
```

Activity 2: Install Visual Studio Code

- Download and install VS Code.
- Explore the VS Code interface.
- Create a Python workspace.

Activity 3: Install Required Extensions

- Install
- Python Extension
- Pylance
- GitHub Copilot
- Jupyter (Optional)

Activity 4: Configure AI Coding Assistants

- Sign in to GitHub Copilot.
- Explore ChatGPT/Gemini.
- Generate your first Python program using AI.

Example Prompt:

Write a Python program to print "Welcome to Python Programming".

Activity 5: Variables

Create variables of different data types.

Example:

```
student_name = "Rahul"
```

```
age = 19
```

```
cgpa = 8.75
```

```
is_hosteller = True
```

Activity 6: Input and Output

Accept user information.

Example

```
name = input("Enter Name: ")  
branch = input("Enter Branch: ")
```

```
print("Welcome", name)  
print("Branch:", branch)
```

Activity 7: Data Types

Explore

Integer

Float

String

Boolean

Use

`type()`

`id()`

to identify data types and storage.

Activity 8: Basic Calculator

Develop a calculator that performs

Addition

Subtraction

Multiplication

Division

Sample Output

Student Utility Calculator

1.Addition

2.Subtraction

3.Multiplication

4.Division

Enter Choice:

Mini Challenge

Enhance the calculator by adding

- Percentage Calculator

- Square

- Cube

- Square Root

- Power Calculation



Thank you